

Python Assignment – Basics of Python Programming

Part A – Print Statements

1. Welcome Message

Write a program to display the following message:

```
Welcome to Python Programming
```

2. Personal Introduction

Create a variable called `name` and store your name in it.

Display:

```
Hello <your name>
```

Example Output:

```
Hello Ayesha
```

Part B – Variables and Arithmetic Operations

3. Addition of Two Numbers

Create two variables:

```
num1 = 25  
num2 = 15
```

Calculate and display their sum.

Expected Output:

```
Sum = 40
```

4. Calculator Program

Store the following values:

```
a = 20  
b = 5
```

Display:

- Addition
- Subtraction
- Multiplication
- Division

Sample Output:

```
Addition = 25  
Subtraction = 15  
Multiplication = 100  
Division = 4.0
```

5. Rectangle Area

Create variables:

```
length = 10  
width = 5
```

Calculate and display the area of the rectangle.

Formula:

```
Area = length × width
```

Part C – Decision Making (if-elif-else)

6. Pass or Fail

Create a variable called `marks`.

Rules:

- Marks 40 or above → Pass
- Marks below 40 → Fail

Display the result.

Example Output:

```
Result: Pass
```

7. Grade Calculator

Create a program using the following grading system:

Marks	Grade
90 and above	A+
80 – 89	A
60 – 79	B
40 – 59	C
Below 40	Fail

Display the grade.

8. Age Category

Create a variable called `age`.

Display:

- Child (0–12)
- Teenager (13–19)
- Adult (20 and above)

Example Output:

```
Category: Teenager
```

Part D – Loops

9. Print Numbers 1 to 20

Use a `for` loop to print numbers from 1 to 20.

10. Print Even Numbers

Use a loop to display all even numbers from 2 to 20.

Output:

```
2
4
6
...
20
```